DE LA SALLE HOLY CROSS COLLEGE

Grade 10 | Date: 11-11-2021

Physical Science | Total: 130 marks

Paper 2 | Time: 2 hours

Examiner: K.Tam | Moderator: A. Kamedza

Question 1:

1.1 D ✓✓
1.2 B✓✓
1.3 B✓✓
1.4 C✓✓
1.5 C✓✓
1.6 A✓✓
1.7 B✓✓
1.8 C✓✓
1.9 C✓✓
1.10 A✓✓

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Question 2

2.1 Atoms of the same element with the same number of protons but differing amounts of neutrons✓✓ (2)

2.2.1 $1s^2 2s^2 2p^6 3s^2 3p^5$ ✓✓ (2)

2.2.2 $35{,}5 = \left(\frac{x}{100}\checkmark\right)35 + \left(\frac{100-x}{100}\checkmark\right)37$

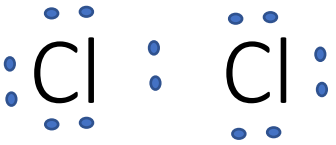
$0{,}35x + 37 - 0{,}37x = 35{,}5$

$1{,}5 = 0{,}02x$

$x = 75\%$✓ (3)

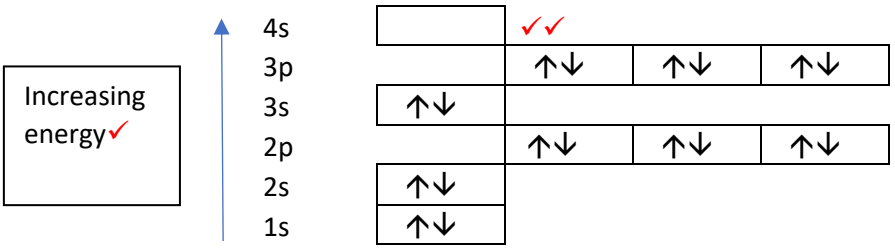
2.3.1 7✓ (1)

2.3.2 (non-polar) covalent✓✓ (2)

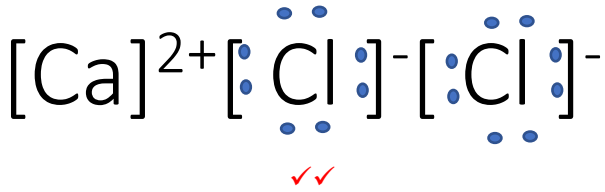
2.3.3 

✓ (1)

2.4.1



(3)

2.4.2 

(2)

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Question 3:

3.1 $CuSO_4$✓ (1)

3.2.1 mass gain (g)✓ (1)

3.2.2 time (s) ✓ (1)

3.2.3 concentration of solutions✓

Mass of electrodes✓

Scale to measure

Powersupply

Temperature (2)

3.3

x-axis✓

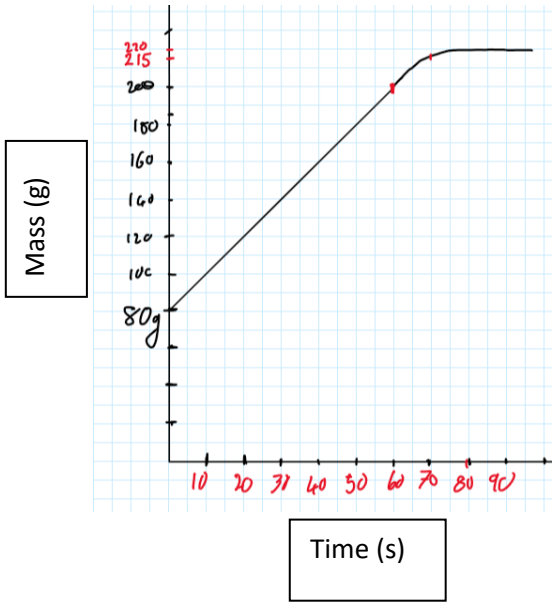
y-axis✓

plotting✓✓

LOBF✓

curve at end✓

heading✓



(7)

3.4 redox reaction✓✓ (2)

3.5 $\frac{\Delta y}{\Delta x} = \frac{(210-80)}{80}$✓✓

$= 1{,}63\ g.s^{-1}$✓ (3)

3.6 The reaction is complete/ finished OR reactants are used up ✓ (1)

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Question 4:

4.1.1 Ionisation reaction✓✓ (2)

4.2.1 $H_2SO_4 + Mg$✓ $\rightarrow MgSO_4 + H_2$✓ (2)

4.2.2 Substance used to test for an acid or a base✓ (2)

4.2.3 hold a lit splint to the opening of the test tube and listen for a pop✓✓

H_2 Gas explodes when exposed to a flame (2)

4.3.1 Acid base reaction/Neutralisation ✓ (1)

4.3.2 $C_{base} = \frac{n_{base}}{v_{base}}$

$n_b = (0{,}05)(0{,}2)$✓

$n_b = 0{,}01\ mol$✓

$n_{acid} = \frac{n_{base}}{2}$✓

$n_a = \frac{0{,}01}{2} = 0{,}005\ mol$✓

$C_{acid} = \frac{n_{acid}}{v_{acid}}$

$C_a = \frac{0{,}005}{0{,}05} = 0{,}1\ mol.dm^{-3}$✓ (5)

4.3.3 potassium sulphate✓ (1)

4.4 $H: \frac{3{,}2}{1} = 3{,}1$

$C: \frac{19{,}4}{12} = 1{,}62$ ✓

$O: \frac{77{,}4}{16} = 4{,}84$

$H:C:O$

$\frac{3{,}1}{1{,}62} : \frac{1{,}62}{1{,}62} : \frac{4{,}84}{1{,}62}$✓

$1{,}9 : 1 : 2{,}9$

$2 : 1 : 3$

$Therefore\ H_2CO_3$✓ (3)

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Question 5:

5.1.1 Measure of the tendency for an atom to attract a pair of bonding electrons ✓✓ (2)

5.1.2 $\Delta EN = 2.1 - 3.0$✓ $= 0.9$✓ (2)

5.1.3 HCl -Difference of EN =0,9✓

-Chlorine has a greater tendency to draw electrons towards itself✓

-chlorine will gain a partial negative charge

Cl_2 -difference of EN is 0✓

- both atoms will have an equal tendency to draw electrons towards themselves✓

-both atoms will have equally shared electrons (4)

5.2.1 Electron affinity is the energy released when an atom gains an electron✓✓ (2)

5.2.2 Ionisation energy is increasing✓ across the period✓ (2)

5.2.3 higher✓ (1)

5.2.4 Sodiums second electron to remove will be from the 2p orbital which will require more energy than magnesium✓. Magnesiums second electron will be in the 3s orbital which will require less energy to remove✓. (2)

Question 6:

6.1.1 HgO✓ (1)

6.1.2 Decomposition✓✓ (2)

6.1.3 $2HgO$✓ $\rightarrow 2Hg$✓ $+ O_2$✓ balance✓ (4)

6.2.1 acid base/ gas forming ✓ (1)

6.2.2 bubble the gas through clear limewater, if the limewater turns milky CO_2 will be present✓✓ (2)

6.3.1 $n = \frac{m}{M}$

$n = \frac{10,6✓}{106✓}$

$n = 0,1\ mol$✓ (3)

6.3.2 $n_{Na2CO3} = n_{H2O}$✓

$m = M.n$

$m = (18 \,.\, 0{,}1)$✓

$m = 1{,}8\ g$✓ (3)

6.3.3 $n_{CO2} = n_{Na2CO3}$✓

$n = \frac{V}{V_m}$

$V = (0{,}1$✓$)(22{,}4$✓$)$

$V = 2{,}24\ dm^3$✓ (4)

6.4.1 $mass_{H2O} = 64{,}6 - 55{,}7$✓

$mass = 9\ g\ of\ H_2O$✓

$n = \frac{9}{18} = 0{,}5\ mol$✓ (3)

6.4.2 $mass_{Na2CO3} = 55{,}7 - 50{,}4 = 5{,}3\ g$✓

$n_{Na2CO3} = \frac{5{,}3}{106}$✓ $= 0{,}05\ mol$✓

$0{,}5\ n_{H2O} : 0{,}05\ n_{Na2CO3}$✓

$10_{H2O}; 1_{Na2CO3}$✓

$Na_2CO_3.10H_2O$ ✓ (6)

Question 7

7.1 precipitation reaction✓✓ (2)

7.2.1 Sulphate✓ or carbonates✓ (2)

7.2.2 White✓ (1)

7.2.3 use dilute nitric acid to determine if a carbonate is present✓. (Fizzing of the CO_2) would be observed and clearing of the solution. ✓ (2)

7.3.1 Bromide✓ ; Br^-✓ (2)

7.3.2 Iodide✓ ; I^-✓ (2)

7.3.3 $2AgNO_3$✓ $+ MgBr_2$✓ $\rightarrow Mg(NO_3)_2$✓ $+ 2AgBr$✓ (4)